

# Darwin Smith

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## EDUCATION

### University of California, San Diego

*B.S. in Mathematics and Computer Science*

San Diego, CA

June 2026

## EXPERIENCE

### Founder, Full-Stack & Applied-AI Engineer

Feb 2026 – Present

*Veraflux ([veraflux.org](https://veraflux.org))*

*San Diego, CA*

- Designed, built, and shipped Veraflux solo: a production platform (React/TypeScript, Node/Express, Python/FastAPI) that turns live clinical research from PubMed into cited, evidence-graded supplement reports personalized to age, sex, and goal, deployed as one Docker unit on Railway with Stripe billing.
- Built the multi-stage LLM pipeline across Gemini and Claude: faceted PubMed searches surface 150+ candidate studies, staged selection keeps the ~50 that hold up, and a verification pass drops any ungrounded citation.
- Kept per-report cost low with per-task model routing, cross-vendor prompt caching, and an incremental refresh that re-synthesizes only when new studies change the findings; reconciled measured cost to the vendor bill to the cent.
- Wrote the backend for production load: Postgres advisory locks for cross-instance concurrency, a multi-layer cache with a refresh state machine, atomic quota accounting, and daily cost caps that bound LLM spend.
- Designed the interface and a from-scratch design system (CSS tokens, dark mode, hand-built icons, a WebGL hero shader), plus a layer that injects interactive citations and glossary definitions into generated prose.
- Made the client-rendered SPA fully crawlable with build-time prerendering, a dynamic sitemap, and schema.org structured data across 200+ programmatic pages.

### Test Engineer

Nov 2023 – Apr 2026

*SiteHive*

*San Diego, CA*

- Managed US-based quality assurance and final validation for IoT hardware deployments, serving as the technical liaison for major enterprise clients.
- Diagnosed firmware and network connectivity faults on Hexanode devices using TeraTerm and STM32CubeProgrammer, isolating and resolving issues before deployment.
- Improved deployment reliability by catching recurring hardware defects and tightening validation procedures.

### Commercial Shuttle Driver

Oct 2023 – Apr 2026

*UCSD*

*La Jolla, CA*

- Drove fixed campus routes 20+ hours a week alongside a full course load, concurrent with the SiteHive role.

## PROJECTS

### Club Triton (Collection Strategy Game) | *JavaScript, IndexedDB, GitHub Actions*

Apr 2025 – June 2025

- Built the core game loop, turn-based battle mechanics, and a deck-building system with persistent inventory in IndexedDB, coordinating cross-team integration through Agile sprints and code reviews.
- Set up CI/CD with GitHub Actions to run builds, ESLint checks, and Playwright regression tests on every push.

### Global Conflict & Defense Stock Analysis | *Python, Pandas, Seaborn*

Sep 2025 – Dec 2025

- Merged 20+ years of daily stock and annual conflict data into a composite index of top US defense contractors.
- Found a strong correlation ( $r = 0.67$ ) between conflict deaths and defense valuations, isolating key confounders.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript/JavaScript, Java, C/C++, SQL, C#, Bash

**Frontend:** React, Framer Motion, WebGL, SVG/CSS

**Backend & Databases:** Node/Express, FastAPI, REST APIs, PostgreSQL (Prisma), Stripe, Supabase, IndexedDB

**Cloud & DevOps:** Docker, Railway, GitHub Actions, AWS, Azure

**AI & Data:** Retrieval-Augmented Generation (RAG), LLM Orchestration, Pandas, NumPy, Scikit-Learn

**Testing & Tooling:** Git, Playwright, Jest, JUnit